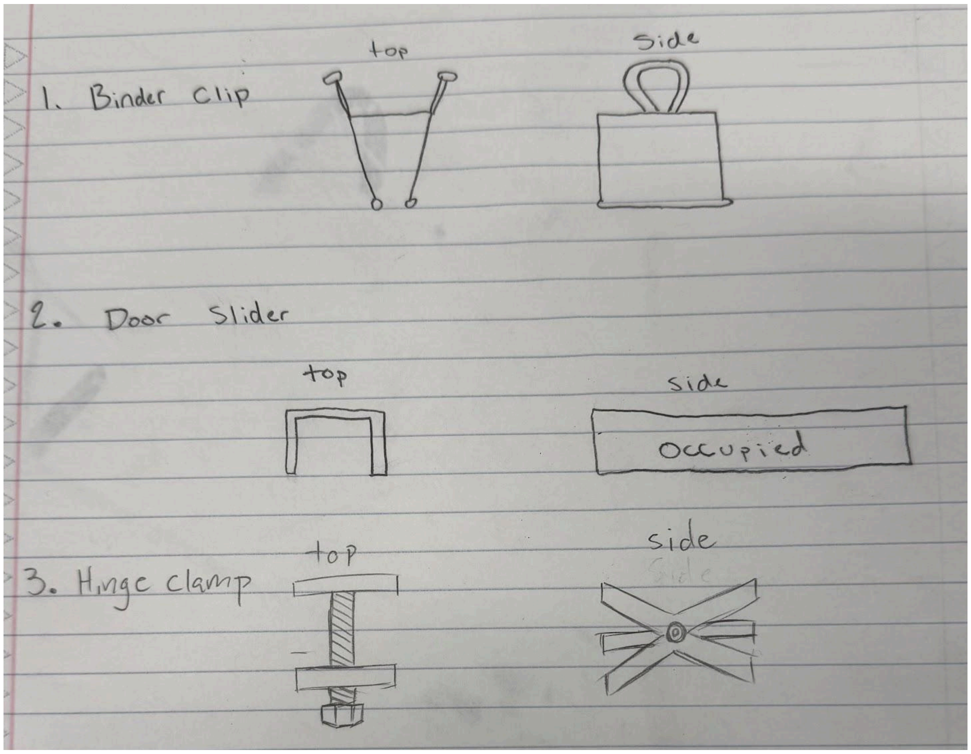
Element D:

Design Concept Generation, Analysis, and Selection



Below is a Decision Matrix to analyze different Prototypes

Criteria	Weight	Binder Clip	Door Slider	Hinge Clamp
Size	.35	2	3	1
Passes TSA	.25	2	3	1
Strength	.15	1	2	3
Degree of Swing	.10	2	1	3
Weight	.07	3	1	2
Ease of use	.05	2	3	1
Safety	.03	2	1	3
SCORE:		1.92	2.45	1.63

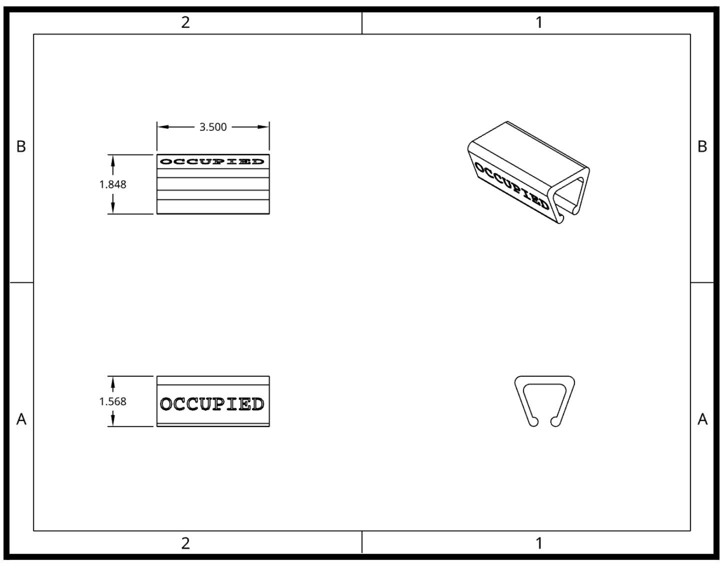
As a group, we each individually made design prototypes, then came together to select 3 of the best prototypes. The 3 design prototypes are displayed above. Among all the prototypes designed, we narrowed our viable solutions down to a binder clip (Solution 1), a door slider (Solution 2), and a hinge clamp (Solution 3). These 3 solutions were the most realistic design solutions since we could use preexisting objects or materials found in the engineering room to construct them. To finalize which design solution we would choose to build/create we use a design matrix. A design matrix gives each criteria, from element C, a weighting based on which is most important to be included in the final solution, and gives each design a score (3-best and 1-worst) which we will add up for a total score. The total score is the sum of each ranking multiplied by criteria weighting. To understand the process explained view the decision matrix displayed below.

Solution 1.)

The Binder Clip design is somewhat self-explanatory. The clip will go on top of the door along the upper rim. It will go between the door of the stall and the wall on the side of the stall. It's made of a durable metal material that will promote longevity. Additionally, it is quite small and strong. In addition, its small size will allow it to be transported easily as it can be placed in the users pocket for easy access.

Solution 2.) - Chosen Design

The Door Slider design will sit on top of the stall door and slide over to be on top of the crack between the door opening and the stall wall. This will be created out of a majority of plastic to help with durability, but will be somewhat large and could potentially cause problems with transportation in the client pocket. However, the simple design will allow for easy and quick access and use.



Solution 3.)

The Hinge clamp design is a screw-like device which will go in between the door crack and each end will fasten tight. This will result in a very slow usability rate as it must be installed in a precise way to utilize the locking function. In addition, the large apparatus will not be easy to transport due to the long rod and bulky perpendicular planks.